

SENTIMENT ANALYSIS

Key Words

1.Sentiment Analysis
2.Natural Language Processing
3.Machine Learning
4.Text Mining
5.Polarity Detection
6.Lexicon-Based Approach
7.Support Vector Machine
8.Naive Bayes
9.Neural Networks
10.Business Intelligence
11.Text Analysis
12.Rule-Based Approach

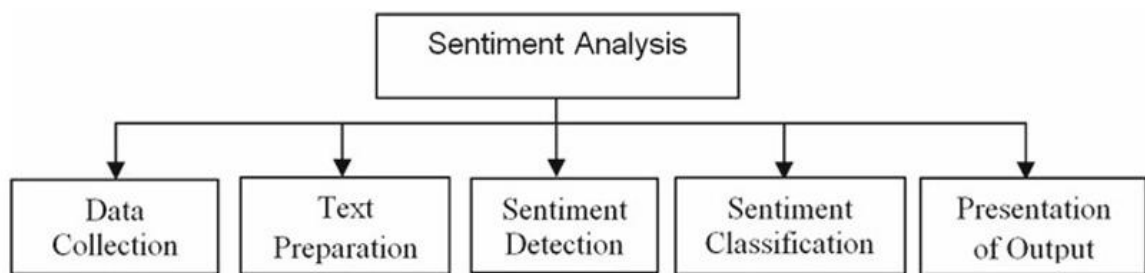
Abstract

The rapid growth of internet-based applications, including social media and blogs, has exploded in user-based comments and reviews. Sentiment Analysis (SA) plays an important role in removing and analysing these opinions to support businesses, governments and researchers in decision making. This study offers a systematic literature review of SA methods, applications and challenges, which emphasizes the role of artificial intelligence in automatic to emotion classification. The SA has employed the Natural Language Processing (NLP) and lesson mining techniques with applications with business, politics and consumer behaviour analysis to classify mining techniques in positive, negative, or neutral emotions. The study also examines large data technologies, especially Hadoop, in processing large -scale emotion data from social networks. Additionally, this machine examines learning and lexicon-based approaches, including innocent Bayes, Maximum Entropy, and the support of vector machines, compare their effectiveness and limitations. Despite the progress, challenges remain in correctly explaining the polarity of emotions and handling linguistic diversity. The purpose of future research is to increase emotional classification accuracy and expand its applications in various domains.

1. Introduction:

Sentiment Analysis, or opinion mining, is one of the natural language processing (NLP) techniques to know the emotion or emotional tone behind a writing. This is the process of conditioning text data in positive, negative or neutral. This technology, which allows companies to better understand the perception of public opinion, customers' response and brand's perception, is rapidly being used in various fields including marketing, customer service, finance and social media monitoring. Digital platforms and the internet have both expanded rapidly, changing the sort of people share their thoughts and experiences. A large portion of user-generated content is concentrated on social media networks, blogs and online review sites which results in an exponential growth of textual data. Everyday millions of customers share comments, reviews and opinions providing business, government entities and research a challenge to make sense out of this unstructured data. One of the most important techniques for analysing and classifying public sentiment is Sentiment Analysis (SA) which assists decision-makers in tracking trends, preferences, and concerns effectively. Sentiment Analysis (opinion mining) is a natural language processing (NLP), text mining and machine learning technique that is used to determine the sentiment expressed in textual data. By assessing feelings, organizations can make data-driven decisions as this process classifies opinions as positive, negative or neutral. SA was first introduced by Nasukawa and has since become a fundamental aspect of artificial intelligence, enhancing automatic sentiment classification across diverse fields.

As digital communication continues to expand, businesses and organizations require robust tools that can quickly analyse massive amounts of text data. You train on data till October 2023. Example: Businesses use sentiment analysis to parse product reviews, measure social media comments, and gauge customer satisfaction. In finance; it is used extensively for stock market trends prediction through news sentiment analysis. Sentiment analysis is also used by governments and politicians to understand public opinions about policies and social issues. Thus, SA has gained wide attention from research communities in recent years, as reflected by the surge in search volumes of related research work and publications. SA has gained more and more scholarly attention in the last ten years. It is the detailed process of how words, phrases and sentences relate to each other in its context, so that machines understand the fine line of human communication. It starts with lexical analysis, examining the individual meanings of words, often utilizing dictionaries and knowledge bases. Next comes compositional semantics, which describes how the meanings of individual words combine to form coherent phrases and sentences. Resolving ambiguities is a necessary step because the same word can have different meanings depending on its surrounding context. Further improving comprehension, semantic analysis typically integrates knowledge graphs and ontologies, which create a framework of real-world knowledge and connections. Algorithms use machine learning techniques that allow models to detect patterns and associations in large datasets, improving predictive capable of identifying context and extracting meaning. In the end, semantic analysis aims to connect human language with how machines understand it. This helps things like search engines, chatbots, and sentiment analysis to work better, allowing them to process and respond to language more accurately and meaningfully. By looking deeper than just the basic structure of sentences, semantic analysis opens up smarter and more natural ways for people and machines to communicate.



Sentiment Analysis

Sentiment analysis is becoming very important to study growing opinions faster and faster with in social media and other sites, The huge explosion in information in recent years in the sites of communication, air traffic and alternative markets, all this huge amount of information cannot be controlled and analysed

used the traditional way so the scientists and researchers developed a high-efficiency techniques to deal with this data. This requires the SA to process data and know its polarity to determine the right decision. SA involves five steps to process data; those are data collection, text preparation, sentiment detection, sentiment classification and presentation of output as shown in bellow (Fig. 1).

Data Collection

The data collection is the first step in sentiment analysis. The collection of data from sources like user groups, Twitter, Facebook, blogs and commercial website such as amazon.com and alibaba.com, etc. This data cannot be analysed using traditional methods like scanning, text analysis, and language processing, which is used for

extraction and classification. We proposed a certain method for a task of paraphrasing and gathering the tweet data called Twitter Streaming.

Text Preparation

Text preparation examines the data before analysing it. Some reviews and conversations in the communication sites contain offensive and inappropriate words, so they are examined and preparation to be the result more reliable analysis. This process selects the contents that are not related to the analysis and then removes it. Objective of the process is the removal of spam and inappropriate reviews before sent to automated analysis. In this case, we can use part of speech (POS) technique which are used for text preparation before analysis.

Sentiment Detection

Sentiment detection is the process of finding the sentiment newline expressed in a review by using machine learning technique or NLP technique; these are also called opinion mining (OM) new line and sentiment analysis. Sentiment detection consists of the examination of phrases and sentences extracted from reviews and ideas. All the sentences containing self-expressions like beliefs, opinions, and abuse are retained. Many research studies in this field included different methods of detection, one of the recent studies propose a system for automatic sentiment detection in natural audio streams by using POS technique.

Sentiment Classification

Sentiment Classification is a task to extraction and classification the text whose objective to classify according to a polarity of the opinion it contains (pang 2002),e.g. positive or negative, good or bad, like or dislike. Sentiment classification contains multiple techniques, and it is classified into three main techniques, namely machine learning approach, hybrid techniques approach, and lexicon-based approach. Presently, Naive Bayes technique and support vector machines (SVMs) are more popular and used for sentiment classification. These techniques can improve an accuracy of classification of Tweets, such as Ankur Goel use Naive Bayes for Sentiment Analysis Tweets in high speed. Therefore, sentiment analysis has a great deal of research in this field and found that many applications and improvement shave occurred in sentiment analysis in recent years. In this study, we will clarification most of the research in this area. These articles cover most of the divisions and classifications widely in SA fields. Sentiment classification techniques are discussed with more emphasis on most details and related points and originating references. In Fig. 2, we will illustrate all techniques which are currently used in sentiment classification until 2017.

Presentation of Output in Sentiment Analysis

The final stage in sentiment analysis is the Presentation of Output, where the analysed sentiment data is visualized and communicated in an understandable format. This step ensures that stakeholders can interpret and act upon the sentiment insights effective.

2. Techniques of Sentiment Analysis

Sentiment analysis helps us understand feelings and opinions in written text. There are several ways to analyse sentiment, and the choice of technique depends on how complicated the text is, what it's being used for, and how detailed the analysis needs to be. The main techniques for sentiment analysis include polarity-based analysis, aspect-based analysis, emotion detection, fine-grained analysis, and intent analysis.

1. Polarity-Based Sentiment Analysis

This is the simplest type of sentiment analysis, sorting text into three groups:

Sentiment	Example
Positive	I love this product.
Negative	The service is bad.

Neutral	The movie was okay.
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People often use this technique in product reviews, comments on social media, and surveys to understand.

2. Aspect-Based Sentiment Analysis (ABSA)

This method takes a closer look by identifying particular features or qualities in the text. For instance, in a review that says:

"The battery life of this phone is great, but the camera quality is poor."

- The battery life is seen as positive.
- The camera quality is viewed as negative.

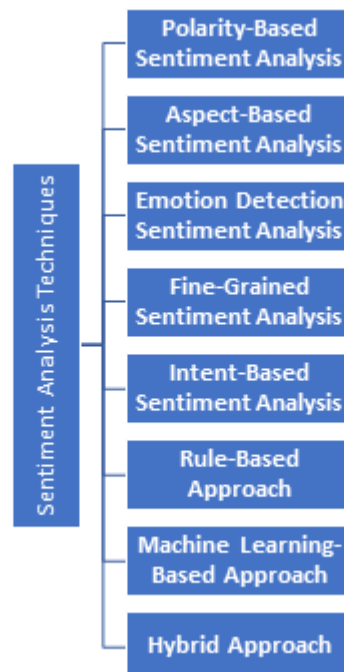
This approach helps businesses analyze customer feedback, allowing them to recognize what they do well and where they need to improve.

3. Emotion Detection Sentiment Analysis

This technique focuses on identifying specific emotions like happiness, anger, sadness, frustration, or excitement expressed in the text. Instead of simply labelling sentiment as positive or negative, it connects it with particular emotions. For example:

Example	Sentiment
I am thrilled with my new job.	Happiness
I am extremely frustrated with the slow internet,	Anger

To carry this out, emotion detection often uses lists of emotions or advanced models that can interpret emotional expressions in writing.



4. Fine-Grained Sentiment Analysis

This technique offers a more detailed way of categorizing sentiment beyond just positive or negative. It usually uses a five-point scale, which can look like this:

Sentiment	Example
Very Positive	Absolutely Amazing.
Positive	I like it.
Neutral	It's okay.
Negative	Not great.
Very Negative	Terrible Experience.

Fine-grained analysis is popular for feedback that includes ratings, like product reviews with star ratings (1-5 stars).

5. Understanding User Sentiment

This method traces sentiments people share and what they actually desire. Take a customer support message for example:

Sentiment	Example	Interpretation
Very Positive	This is the best product I have ever tried.	Strong positive emotion, highly satisfied.
Positive	I like it, works well for me.	Generally happy with experience.
Neutral	It's okay, nothing special.	Mixed feelings.
Negative	Not great, expected better.	Disappointed.
Very Negative	Worst experience ever.	Strong negative emotion.

It indicates the person is not happy but also indicates they want support. You will see such analysis in chatbots, customer service tools, and online shopping websites. It's useful to know whether someone cares, is demanding specifics or wants a solution.

6.Rule-Based Method

Use rules and word lists to parse text.

It assigns scores to words, and computes the overall sentiment based on those scores.

For example, a rule-based system would treat "great service" as positive and "terrible experience" as negative.

Sentence	Sentiment
I love this phone, it's amazing.	Positive
This product is terrible and useless.	Negative
The movie was just okay.	Neutral

But it suffers from a lack of understanding of context and informal language.

7.Machine Learning Method

It utilizes several algorithms like Naïve Bayes, support vector machine (SVM), deep learning for sentiment identification.

Sentiment models are trained on labelled data to detect trends.

So, a deep learning model can classify if the reviews on the movie is positive or negative given the data it has seen. Recurrent neural networks (RNNs) and transformers (e.g. BERT) that significantly improve the accuracy belong to it.

Sentence	Model Prediction	Confidence Score
I absolutely love this phone.	Positive	95%
This is the worst service ever.	Negative	97%
The movie was fine.	Neutral	85%

8. Blended Approach

This method combines rule-based and machine learning techniques for better results.

It combines language rules with machine learning models, to classify sentiment better

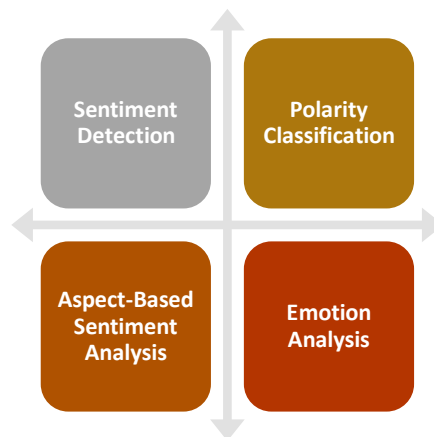
For instance, a chatbot analysing customer feedback could combine predefined rules with prediction models for a more nuanced insight.

Sentence	Rule Based Check	Final Sentiment
Absolutely fantastic product.	Positive	Very Positive
I hate this phone, it's the worst.	Negative	Very Negative
It's okay, nothing special.	Neutral	Neutral

3. Core Tasks in Sentiment Analysis:

Sentiment analysis is a complex task that encompasses various essential components and applications. Below are some of the principal tasks and uses connected with sentiment analysis:

1. Sentiment Detection: This task focuses on determining whether a given piece of text conveys a positive, negative, or neutral sentiment. Various methods, including lexicon-based approaches and machine learning models, are commonly utilized for this purpose.

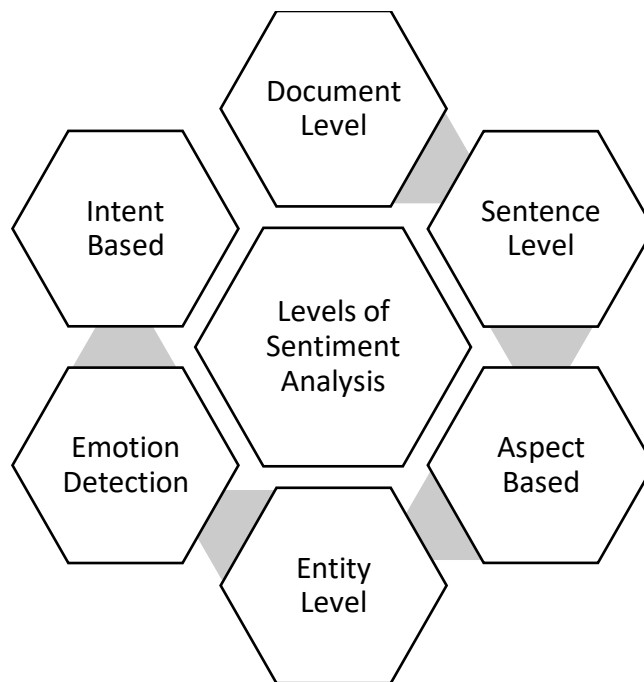


2. Polarity Classification: This specific task classifies the sentiment expressed in the text into distinct levels of positivity or negativity. Often represented as a sentiment score, it quantifies sentiment on a numerical continuum.

3. Aspect-Based Sentiment Analysis (ABSA): ABSA represents a more nuanced form of sentiment analysis, concentrating on particular aspects or features of a product or service. It identifies sentiments linked to specific elements, such as quality, service, or atmosphere.

4. Emotion Analysis: This process involves distilling sentiment into more precise emotions, such as happiness, excitement, or trust. It allows for a richer comprehension of the emotional undertone present in the text. Incorporating these tasks improves the granularity of understanding in sentiment analysis, offering profound insights into consumer opinions and emotional responses.

4. Levels of Sentiment Analysis:



1. Document Level Sentiment Analysis:

Document-level sentiment analysis is about determining general meanings-it is positive, negative, or neutral-as a customer review, a news article or a complete document of social media posts. This works on estimates that the document expresses a major feeling.

How it works:

Input text: You start with the entire document, it is a review, article or email.

Preprocessing: After that, you prepare the text by eliminating surplus words and punctuation.

Feature extraction: This process converts the text into numbers, using techniques such as TF-Eid or Word Embeddings.

Emotion classification:

- Lexicon-based methods: They use a sentiment dictionary such as sentiments or warders.
- Machine Learning Model: This includes classifier such as LSTMs and Transformers such as LSTMs and deep learning models.

Output: Finally, the entire document receives a label in the form of positive, negative or neutral.

Example:

Review: "The hotel was amazing! People were good, and food was very good."

Result: Positive

Review: "The service was terrible. It took me an hour to get my food, and it was cold."

Result: Negative

Use cases:

- Analysis of product reviews on e-commerce platforms.
- To review the news headlines to see how public sentiment changes.
- Analysis of customer response to improve professional services.

2.Sentence Level Sentiment Analysis:

A sentence-tiered sentiment analysis determines the spirit of each individual sentence in a text. This approach is useful when a document has mixed feelings (both positive and negative statements).

How it works:

Input text: A paragraph, review or social media posts are given as inputs.

Sentence tokening: The text is divided into individual sentences.

Preprocessing: Text cleaning (removal of stop words, punctuation, etc.).

Emotion classification for each sentence:

Lexicon-based (eg, warder, centovardnet)

Machine learning-based (Bhole Bayes, SVM, Transformer Model)

Deep Learning-based (LSTM, Bert, GPT)

Output: Each sentence is labelled as positive, negative or neutral.

Example

Review:

"The phone has a great camera. Battery life is disappointing. Overall, it is an average product."

Sentence-level feelings:

"The phone has a great camera." → positive

"Battery life is disappointing." → negative

"Overall, it is an average product." → neutral

Use cases:

Social media analysis: detecting emotions mixed in a tweet or post.

Customer Review: Identify both strengths and weaknesses in a product review.

Survey reactions: Analyse the detailed response from users.

3.Aspect Based Sentiment Analysis:

Aspect-based sentiment analysis (ABSA) is an advanced form of emotion analysis that identifies expression expression for a product, service or unique specific aspects (features, characteristics, or components).

How it works

Input text: review, feed or collect any text data.

Aspect extraction: Identify major aspects or institutions in the text (eg, "camera," battery, "" service ").

Sentiment classification: Set emotion for each aspect individually.

- Positive
- Negative
- Neutral

Output: A structured representation of aspects and their respective feelings.

Example of ABSA:

Review:

"The quality of the camera is amazing, but the battery life is bad."

<u>Aspect</u>	<u>Sentiment</u>
Camera	Positive
Battery	Negative

Methods used in ABSA:

- Lexicon-based methods: Use predetermined emotion dictionaries (eg, cantwardate, warder) to analyse the spirit of words associated with aspects.
- Machine Learning-based methods: train models to classify emotions for each aspect (good Beyas, SVM).
- Deep Learning-based methods: Use advanced NLP model (LSTM, Burt, Transformer Model) to remove aspects and determine the soul.

Use Cases of ABSA:

- E-commerce and product reviews→ Identify customers' opinions on various characteristics (eg, price, design, durability).
- Hotel and Restaurant Review →Analyse the reaction of food, service, environment, hygiene.
- Social media monitoring →Understand customer spirit on specific brand features.

4.Entity-level Sentiment analysis:

Unit-level sentiment analysis is a more granular approach to emotion analysis that focuses on detecting emotions associated with specific institutions mentioned in a lesson (eg people, brands, products, places or organizations).

How it works

Input lesson: Text containing one or more institutions (eg, a product review, tweet, news article).

Nominated unit recognition (NER): Recognize institutions (eg, "apple," "Tesla," "New York").

Sentiment classification: Assign each unit a feeling:

- Positive
- negative
- neutral

Output: Structured representation of institutions with their related feelings.

Example of Entity Level Sentiment Analysis:

Review:

"I love Apple's new iPhone, but Samsung's latest model is disappointing."

<u>Entity</u>	<u>Sentiment</u>
Apple	Positive
Samsung	Negative

Methods Used in Entity Level Sentiment Analysis:

- Named Entity Recognition(NER)→identifies institutions using NLP models such as Spacy, NLTK, or transformer-based model.
- Lexicon-based methods→ Each uses emotion dictionaries to determine emotion polarity for words associated with each unit.
- Machine learning and deep learning→ uses models such as SVM, LSTMS, or Bert to classify emotion for each unit.

Use cases of Entity Level Sentiment analysis:

- Brand Monitoring→ Analyse the customer's opinion on many brands (eg, Apple vs Samsung).
- Political Sentiment analysis→ Determine public opinion on politicians or parties.
- Financial Market Analysis→Assess the feeling of companies, stock or cryptocurrency.

5.Emotion Detection Sentiment Analysis:

The emotion detection sentiment analysis is beyond the basic positive, negative and neutral emotion classification, which to identify specific feelings expressed in a lesson, such as happiness, anger, sadness, fear, surprise, hatred, etc.

How it works:

Input text: A sentence, review, tweet or any text data.

Preprocessing: Cleaning the text (stop word, removal of punctuation, etc.).

Feature Extraction: Converting text into numerical form

Example: TF-Eid, Word Embeddings, or Transformer-based embedding.

Emotion classification: Using models to detect emotions from text:

- Lexicon-based methods: Uses predetermined emotion dictionaries (eg, NRC emotion lexicon).
- Machine learning methods: Use classifier such as Bhole Bayes, SVM, or Deep Learning Model (LSTM, Burt).

Output: Provides the text one or several emotions.

Examples of emotion detection

Review:

"I am very excited about my new job, but I am a little nervous too!"

<u>Emotion</u>	<u>Detected</u>
Joy	Yes
Nervousness	Yes
Sadness	No

Common emotions detected in Text:

- Joy/Happiness → "I had a wonderful time in the concert!"
- anger → "Customer service was terrible!"
- Sadness → "I feel very lonely these days."
- Fear → "I am worried about my exam results."
- Surprise → "Wow! I didn't expect this to happen."
- Hatred → "This place is very dirty and untoward."

Methods used in emotion detection

- Lexicon-based: NRC Emotion uses predetermined emotional words such as Lexicon, Word-Aspect.
- The machine uses a trained model on the dataset labelled to classify the learning-based → emotions.
- Deep learning-based: Uses berts such as lstms, cnns, or transformer-based models for high accuracy.

Use cases of detection of emotions:

- Social media analysis: Understanding emotional trends in posts and comments.
- Customer response analysis: Detect emotions in reviews for improvement in products/services.
- Mental health analysis: Identifying emotional patterns in interaction or medical sessions.
- Chatbots and Virtual Assistant: Increase Chatbot Response based on user's feelings.

6.Intention-based emotion analysis

Intention-based emotion analysis is beyond a statement to understand the user's intentions and identify positive, negative or neutral emotion. It helps determine whether there is a user:

- Asking for information
- Help Request
- Making a Complaint
- Appreciate
- Expressing Sarcasm

It is useful in customer aid, chatbots and business intelligence what users expect action.

How it works:

Input text: A sentence, review, or user message is provided.

Clean the preprocessing → text (stop words, remove punctuation).

Intention recognition → Classes the text based on preceding intentions categories:

- Request for help → "Can you help me with my order?"
- Complaint → "Service was terrible!"
- Appreciation → "This product is amazing!"
- Inquiry → "How do I reset my password?"
- Suggestion → "You should improve battery life."

Sentiment analysis: determine whether the intention is expressed positive, negative or neutrally.

Output: Both intentions and Bhavna.

Example of intention-based Sentiment Analysis

User Messages:

"I like this phone very much, but the batteries are very fast drains. Can you help?"

<u>Intent</u>	<u>Sentiment</u>
Praise	Positive
Complaint	Negative
Request for Help	Neutral

Methods Used in Intent Based Sentiment Analysis:

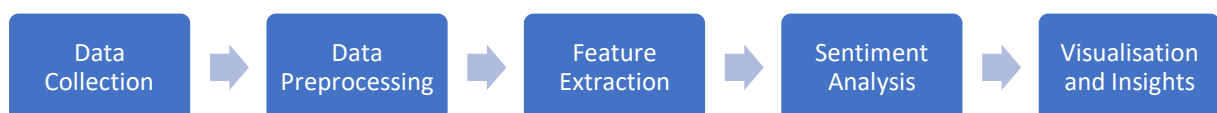
- Rules-based methods → Use predetermined rules to detect intentions based on keywords.
- Machine Learning Model → Train Classifier (SVM, Random Forest, Bhole Bayes) on intention data.
- Deep Learning Model → Use LSTMS, CNN, or transformer (bert, GPT) for high accuracy in detection.

Use Cases of intention-based sentiment analysis

- Find the user's intentions for chatbots and virtual assistant → better reactions.
- Customer support automation → prefer immediate requests based on emotion.
- Social media and brand monitoring → Understand what users are praising, complaining, or seeking help.
- E-commerce and review → Identify customer needs and improve recommendations.

5.Methodology Used for Sentiment Analysis

The functioning of sentiment analysis includes several key steps from data collection to interpretation of results. Below is a structured approach to analyse the sentiment analysis:



1. Problem Definition:

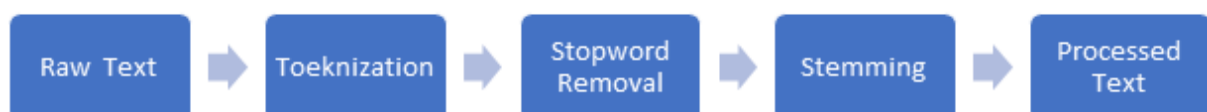
- Define the objective (eg, analysing the customer response, social media sentiment, or product reviews).
- Determine scope (eg, polarity detection, emotion classification, aspect-based emotion analysis).

2. Data Collection:



- Source: Social Media (Twitter, Facebook), Customer Review (Amazon, Yelp), News Articles, or Survey.
- Methods:
 - 1.Web scrapping (eg, using beautifulsoup, scrapy).
 - 2.API (eg, Twitter API, Google Review API)
 - 3.Pre-existing Datasets (eg, IMDB reviews)

3. Data preprocessing:



Text cleaning:

- Remove stopwords, punctuation marks, special characters and numbers.
- Convert the text to lowercase.
- Handle emojis and slang (eg, "😂" → "laughing").

Tokenification:

- Split the text into individual words (tokens) using NLTK or Spacy.

Stemming and lemmatization:

- Reduce words in their root forms (eg, "running" → "run").

Handling negation:

- Change "not happy" in "not_happy" to maintain sentiment.

Example:

Input: "I really love this phone! It's amazing.!!!"

Processed Output: ["Love", "Phone", "Amazing"]

4. Feature extraction:



Bag of words (bow):

- Converts the text into a vector of word counts.

Term Frequency-Invers Document Frequency (TF -IDF):

- Measure the importance of words in a document.

Word embedding:

- Word2Vec, GloVe, or FastText to capture the relevant meaning.

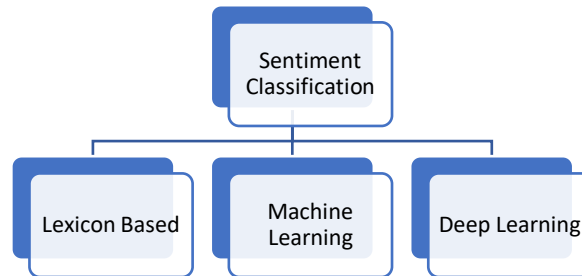
Sentiment lexicon:

- Use dictionaries like VADER, AFINN, SENTIWORDNET to assign sentiment score.

Example:

Word	TF-IDF
Love	0.56
Phone	0.42
Amazing	0.71

5. Sentiment Classification Methods:



Lexicon Based Approach:

- Use lexicon and predetermined rules to determine the sentiment.

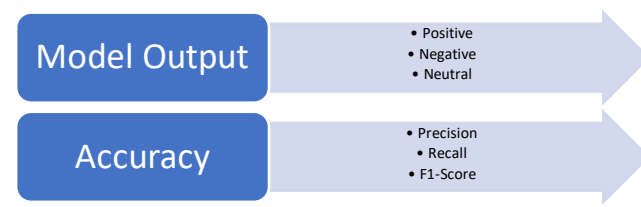
Machine learning approach:

- **Supervised:** Train Classifier (eg, Naive Bayes, SVM, Random Forest, Logistic Regression) on labelled sentiment data.
- **Unsupervised:** Cluster similar sentiments using K-Means, LDA

Deep learning approach:

- Recurrent Neural Network (RNNs), LSTMS, GRU for sequence learning.
- Transformer (BERT, GPT) for contextual sentiment understanding.

6. Model Evaluation:



Performance Metrics:

- Classification for Accuracy, Precision, Recall, F1-score.
- Mean Squared Error (MSE) if using regression-based sentiment scoring.

Cross Validation:

- Use K-fold cross-validation to ensure robustness.

7. Interpretation and Visualisation:

Sentiment Trend:

- Plot sentiment over time using matplotlib, seaborn, or power bi
- Highlight frequently used positive/negative words.

Word cloud:

Aspect-Based Sentiment Analysis (ABSA):

- Identify the sentiment towards specific product aspects.

Sentiment Distribution Chart:

Positive  70%

Negative  20%

Neutral  10%

8. Deployment and Automation:

Deploy Models:

- Use Flask, FastAPI or cloud services (AWS, GCP, Azure).

Automate Sentiment Analysis:

- Implement in real time dashboards, chatbots or CRM systems.

6. Need For Sentiment Analysis:

The Sentiment Analysis (also known as opinion mining) is important in various fields to understand the feelings, ideas and attitudes expressed in the text. Why emotion analysis is important here:

1. Business and Marketing Insight:

- Brand Monitoring - Understand how customers feel about a brand, product or service.
- Competitive Analysis - Analyse public sentiment towards competitors to improve strategies.
- Product improvement - Gain insights from customer feedback and reviews to enhance the quality of the product.

Example:

- A company analyses customer reviews that many users complain about battery life. This helps them improve future product design.

2. Customer Experience and Support

- Real-time Customer Feedback - Analyse customer satisfaction levels based on online reviews and social media posts.
- Automatic Customer Support - Chatbots with sentiment analysis can prioritize anger or dissatisfied customers for rapid response.
- Improve customer loyalty - Identify and resolve negative sentiment trends to increase customer retention.

Example:

- An airline uses sentiment analysis on Twitter to detect negative feedback and quickly respond to unhappy travellers.

3. Financial and Stock Market Analysis

- Investor Sentiment Analysis - Track the public sentiment about companies and stocks to predict market movements.
- News and Social Media Monitoring – Analyse Financial News and Social Media Aviation to make informed business decisions.
- Risk management - Find out possible crises based on negative trends in financial reports.

Example:

- A hedge fund uses an AI-managing sentiment analysis to predict stock price movements based on financial news and social media discussions.

4. Politics and public opinion

- Election prediction - Analyse public sentiment towards political candidates before elections.
- Policy Evaluation - Assess public reaction to government policies and reforms.
- Crisis Management - Detect negative sentiment trends to take proactive action during political crises.

Example:

- Before an election, political parties analyse tweets and news sentiment to adjust their campaign strategies.

5. Health Services and Mental Health

- Mental Health Monitoring – Detect emotional distress or depression from social media posts.
- Patient feedback analysis - improve health services by analysing hospitals and patient response.
- Public Health Awareness - Track the public sentiment about diseases, vaccines and medical remedies.

Example:

- During an pandemic, sentiment analysis helps track misinformation about public concerns and vaccines.

6. E-commerce and Online Reviews

- Product Recommendations - Personalize product suggestions on the basis of customer sentiment.
- Review filtering - Identify fake reviews by analysing the stability of sentiment.
- Improved Search Result-Sentiment-manufacturing relevance based on rank search results.

Example:

- Amazon uses sentiment analysis to rank and highlight the most helpful customer reviews.

7. Legal and compliance

- Detection of abusive language - Identify and flag abusive, toxic or inappropriate content.
- Fake news detection - Analysis sentiment trends to detect misinformation and propaganda.
- Regulatory Compliance - Monitor online discussions for compliance violations in industries like banking and finance.

Example:

- Social media platform uses sentiment analysis to detect and remove abusive language and harmful content.

Conclusion:

Sentiment Analysis is a powerful tool for businesses, governments and organizations to understand human emotions on a scale. It helps in decision making, trend prediction and improvement of customer experiences.

7.Tools Used For Sentiment Analysis:

<u>Tool</u>	<u>Description</u>
NLTK	Python library for NLP, includes Sentiment Analysis
TextBlob	Simple python tool for text processing and sentiment scoring.
VADER	Lexicon based sentiment analyser for short texts like tweets.
Scikit-Learn	ML based sentiment classification using naive bayes, SVM.
Tensorflow and PyTorch	Used for deep learning-based sentiment analysis (LSTM, BERT)
Spacy	Fast NLP library, used with word embeddings for sentiment analysis.
Hugging Face Transformers	Pre trained deep learning models (BERT, GPT) for sentiment analysis.

8.Evaluation Metrics for Sentiment Analysis:

Evaluating a sentiment analysis model ensures it's effectiveness in accurately classifying text as positive ,negative or neutral. The following are the key metrics used to evaluate sentiment analysis models.

1.Accuracy:

Measures the percentage of correctly classified sentiments.

Best for balanced datasets where each class has an equal number of samples.

Formula:

$$\text{Accuracy} = \frac{TP+TN}{TP+TN+FP+FN}$$

- TP (True Positive): Correctly predicted positive sentiment.
- TN (True Negative): Correctly predicted negative sentiment.
- FP (False Positive): Incorrectly predicted positive sentiment.
- FN (False Negative): Incorrectly predicted negative sentiment.

Example:

If a model correctly classifies 90 out of 100 reviews, the accuracy is 90%.

2.Precision:

Measures the proportion of correctly predicted positive statements out of all predicted positive cases.

Useful when false positives are costly.

Formula:

$$\text{Precision} = \frac{TP}{TP+FP}$$

Example:

A model predicts 100 reviews as positive, but only 80 are actually positive.

$$\text{Precision} = 80/100 = 0.8$$

3.Recall (Sensitivity):

Measures the proportion of correctly identified positive statements out of all actual positive cases.

Useful when false negatives are costly.

Formula:

$$\text{Recall} = \frac{TP}{TP+FN}$$

Example:

There are 100 positive reviews, but the model detects only 70.

$$\text{Recall} = 70/100 = 0.7$$

4.F1-Score:

Harmonic mean of Precision and Recall.

Balances precision and recall, especially for imbalanced datasets.

Formula:

$$F1-Score = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

Example:

If precision is 80% and Recall is 70% then

F1-Score=0.746

5. Confusion Matrix:

A table that shows the number of correct and incorrect positions.

	Predicted Positive	Predicted Negative
Actual Positive	TP (50)	FN (10)
Actual Negative	FP (5)	TN (35)

6. ROC Curve and AUC Score:

ROC (Receiver Operating Characteristic) Curve: Plots True Positive Rate (PTR) vs False Positive Rate (FPR).

AUC (Area Under Curve): Measures the area under the ROC curve. Higher values indicate better performance.

Range of AUC Scores:

AUC Score	Performance
0.9-1.0	Excellent
0.8-0.9	Good
0.7-0.8	Fair
0.6-0.7	Poor
0.5-0.6	Fail

Example:

If AUC Score is 0.85, the model performs well in differentiating sentiments.

7. Mean Squared Error for Regression Models:

Used when sentiment score is continuous. (e.g. ranging from -1 to 1 instead of categories)

Formula:

$$MSE = \frac{1}{n} \sum (x - y)^2$$

x=Actual Sentiment Score

y=Predicted Sentiment Score

Example:

If actual sentiment is 0.8, but the model predicts 0.5, the error is higher.

Metric	Use Case	Best For
Accuracy	Overall Performance	Balanced Datasets
Precision	Avoiding Fake Positives	Customer Feedback, Fraud Detection
Recall	Avoiding Fake Negatives	Detecting complaints
F1-Score	Balance between Precision and Recall	Imbalanced Datasets
Confusion Matrix	Understanding model errors	Error Analysis
AUC-ROC	Evaluating classification quality	Multi class sentiment analysis
MSE	Continuous sentiment prediction	Regression base sentiment models

9. Datasets:

1. <https://www.kaggle.com/datasets/jp797498e/twitter-entity-sentiment-analysis>
2. <https://www.kaggle.com/datasets/mdismielhossenabir/sentiment-analysis>
3. <https://www.kaggle.com/datasets/sbhatti/financial-sentiment-analysis>
4. <https://www.kaggle.com/datasets/ankurzing/sentiment-analysis-for-financial-news>
5. <https://www.kaggle.com/datasets/bittlingmayer/amazonreviews>
6. <https://www.kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health>
7. <https://www.kaggle.com/datasets/yasserh/imdb-movie-ratings-sentiment-analysis>
8. <https://www.kaggle.com/datasets/clovisdalmolinvieira/news-sentiment-analysis>
9. <https://www.kaggle.com/datasets/yasserh/imdb-movie-ratings-sentiment-analysis>
10. <https://www.kaggle.com/datasets/chaudharyanshul/airline-reviews>

11. Google Colab Link

<https://colab.research.google.com/drive/15z3KvGuNdwg2sAupiBytHh2hClFcqxXY?usp=sharing>

12. Conclusion:

The feelings, thoughts and feelings expressed in written words have become a very important tool for understanding human feelings and attitudes, which is why the emotion analysis has actually stopped. By taking advantage of natural language processing (NLP), machine learning and deep learning techniques, emotion analysis has found extensive applications in business, marketing, finance, politics, healthcare and social media surveillance. The study highlights various emotion analysis approaches, including various emotions, machine learning and hybrid methods, while also discusses emotional classification techniques such as documents, sentences and aspect-based analysis at various levels. While technology is really advanced and we are very good through emotion from different types of statements - Email, status updates - you know, still really really a big challenge to manage satire and accessories. And this is also especially difficult when we are working with language changes because sometimes the meaning of words can be completely different when different or negative styles are used depending on reference. This is a difficult business that is sometimes right. Future research should focus on enhancing classification accuracy, integrating multilingual abilities and improving relevant understanding to make emotion analysis more strong and effective in various domains. As the interaction is more electrified by digital means, emotion analysis matters a complete. We use it to make a really good decisions, to ensure that customers prefer to interact with us, and develop amazing new bright equipment for analysis based on artificial intelligence.